Main Experimental Ladder: V1 to V5-B

The no-pretrain path was an experiment chain: each version changed one thing, measured it, then exposed the next bottleneck.

Step-by-Step Story

Each version was a controlled answer to the previous failure, not a random architecture swap.

V1

Why: First prove the task was learnable from synthetic labels.

Changed: Scratch CNN at 224 px with basic diagnostics.

Learned: There was learnable signal, but the model was fragile.

V2

Why: If V1 was capacity/data-limited, improve both before blaming transfer.

Changed: Full QC labels, 384 px input, residual CNN, targeted synthetic scale-up.

Learned: Synthetic results improved; the real-domain gap stayed visible.

V3

Why: Test whether the gap was caused by training setup rather than data.

Changed: Searched augmentation, normalization, samplers, and ordinal/coarse variants.

Learned: Low augmentation helped; GroupNorm/ordinal-coarse did not solve transfer.

V3.1

Why: Make the synthetic validation split less contaminated by prompt similarity.

Changed: Moved to metadata_family_holdout by prompt/image family.

Learned: Evaluation became harsher and more honest.

V4

Why: Stop simplifying the target: train the real 10 severity bands directly.

Changed: Used score_band as the target with soft ordinal labels.

Learned: Near-band synthetic behavior improved; real generalization was still weak.

V4.5

Why: Check whether the loss should punish far-away severity mistakes more.

Changed: Added distance-aware EMD-style loss on the 10-band distribution.

Learned: It gave small real gains, but not a decisive fix.

V5-A

Why: Ask whether severity is better represented as one ordered scalar.

Changed: Regressed normalized representative score, then mapped back to bands.

Learned: Real mean-bias improved, but synthetic exact-band accuracy fell.

V5-B

Why: Combine the useful signals instead of choosing one target form.

Changed: Multitask scalar + 10-band + coarse supervision.

Learned: Tolerance on real images widened; longer training showed scratch had plateaued.

Takeaway

The ladder shows why pretrained is a comparison point, not the whole project: V1-V5-B proved the synthetic task was learnable, then narrowed the unresolved problem to real-domain transfer under a scratch CNN.